

MF-07: Lean verification of the uniform product bound

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Scope. This is a formalization and verification note for the complete original MF-07 target. The mathematical resolution remains credited to Matthew J. Colbrook, Department of Applied Mathematics and Theoretical Physics, University of Cambridge. His shared manuscript, *Uniform polynomial product bounds and sharp Hölder continuity of the joint spectral radius*, Theorem 1 and Proposition 5, is retained unchanged in the repository. Epperlein and Wirth retain credit for the original question.

Original target and formal result

For each positive integer d , there is a constant $\Theta_d > 0$, chosen before the family, such that every nonempty compact $\mathcal{M} \subset \mathbb{C}^{d \times d}$ of joint spectral radius one satisfies

$$\|A_k \cdots A_1\|_2 \leq \Theta_d (Lk)^{d-1}, \quad L = \max_{A \in \mathcal{M}} \|A\|_2,$$

for every $k \geq 1$ and every $A_1, \dots, A_k \in \mathcal{M}$. The norm is the Euclidean operator norm and the radius is the original limit of the maximal product norm to the power $1/k$.

The final Lean declaration is `NLA.MF07.canonical_uniform_bound`. The declaration `NLA.MF07.radius_one_semantics` proves the required infimum/root-limit correspondence. Infinite compact families, reducibility, repeated factors, arbitrary complex entries and $d = 1$ are included. The formal proof takes

$$\Theta_1 = 1, \quad \Theta_d = d \left(\frac{6d^2}{d-1} \right)^{d-1} \quad (d \geq 2).$$

This proves the original existential assertion. The manuscript's smaller constant with $2e$ in place of 6, sharpness, general-radius formula, bounded/real-family extensions and separate MF-05 result are outside this formalization. No irreducibility or assumed exact extremal norm is added.

Formalization appendix: executed verification

The immutable proof revision is

b2b9acc83ad7d6b7d3888de8390475a30c720af8.

Actual GitHub Linux run 35172783207, verify job 105047879291, completed successfully on 17 September 2026 (UTC). All 297 submitted inputs and all 18 exported statements passed default-kernel replay, Lean Comparator, permitted transitive-axiom checks and the actual per-project rejection and isolation controls. The verify process and sandbox used non-root UID 1001. The separate global checker-control job was skipped; the required controls ran inside this successful proof job.

Only `propext`, `Classical.choice` and `Quot.sound` occur in the 18 exported axiom lists. The single numerical calculation, $\exp(1) \leq 3$, uses LeanCert kernel trust and is consumed by the final dimension-only constant. All matrix, dimension, length and threshold estimates are symbolic, with no interval subdivision.

The project pins Lean 4.33.1, Mathlib revision 0df444a360eaa60ab8c11dca51a86af692955474 and LeanCert revision 621a43d7cf21f87872392a01e874f2f1dbddc926.

From `matrix-functions-and-stability/MF-07/lean`, run `lake build` with the pinned dependencies. The project README supplies the full non-root Linux harness command and retained evidence.

Immutable proof source • Actual Linux verification run.

Review and publication limits. Two nonauthor agents reviewed the complete proof source. An independent agent and the coordinator authenticated the actual GitHub evidence. These audits are not additional compiler runs or claims that the checker or platform is infallible. The earlier macOS local development is a distinct execution. Later publication-commit and upstream-PR checks remain separate from the immutable proof run above.

Attribution. George Stepaniants supplied the AI-assisted formalization and verification submission with substantial OpenAI Codex assistance. Original mathematical authorship is retained. Proof authors and independent referees have explicit separate roles. No official Tau Ceti endorsement, external human peer review or new historical-priority claim is asserted.